Dynamics of Non-Local Rope Correlations

A Partial Result: How Far the Rope Ontology Reproduces Quantum Correlations, and the Exact Step That Remains

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Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it.

Abstract

The quantum-foundations paper established that the rope ontology, if it describes entanglement at all, must be a NON-LOCAL hidden-variable theory, and deferred the central calculation: can rope dynamics reproduce the singlet correlation $E(a,b) = -\cos(a-b)$, saturating the Tsirelson bound $2\sqrt{2}$, while preserving no-signalling? This paper attempts that calculation and reports a PARTIAL result honestly. We do not claim to have derived quantum mechanics from ropes; we claim to have localised, precisely, the one step on which the question turns. Three things close cleanly. First, a local shared-orientation rope model gives a triangle-wave correlation capped at CHSH = 2 — falsified, exactly as Bell requires. Second, a non-local update model, in which measurement at one end updates the joint two-strand configuration, reproduces the cosine and reaches Tsirelson (numerically 2.83). Third, that model preserves no-signalling exactly: the non-locality lives in the correlations only, never the marginals. One thing does NOT close: the non-local model reaches the quantum value only when the detector couples to the Hopf BASE-sphere (Bloch) angle rather than the spinor fibre angle. The rope geometry supplies both angles; which one the measurement dynamics select is not yet forced by the ontology. We show the entire viability question reduces to this single angle-identification, which is a sharper and more useful outcome than either an unearned success or a vague failure.

The result in one paragraph

CLOSES: (i) local rope model falsified (CHSH ≤ 2, triangle wave); (ii) a non-local joint-update model reproduces $-\cos(a-b)$ and saturates Tsirelson (2.83); (iii) no-signalling holds exactly for the update rule.

DOES NOT CLOSE: the quantum value requires the detector to couple to the Bloch (base-sphere) angle with a 1:1 map; the rope dynamics do not yet FORCE this choice over the competing spinor-fibre angle. The viability of the rope model as a theory of quantum correlations reduces entirely to this one identification. We state it as open, not solved.

1. The Deferred Calculation

The quantum-foundations paper reached a forced conclusion: if the rope model describes entangled pairs and the imbalance is real and definite, then it is a hidden-variable theory; Bell plus the loophole-free experiments rule out the LOCAL version; and retaining realism and measurement-independence leaves only a NON-LOCAL hidden-variable theory. That paper was careful to say it had identified the required ontology but did NOT possess the dynamical law, and it named the missing piece: a mechanism that reproduces the cosine correlation at the Tsirelson bound while preserving no-signalling. This paper attempts exactly that, and reports what happens.

We follow the programme's established method: compute first, and report the result even when it is a partial one. As it turns out, the honest outcome here is neither success nor failure but a sharp LOCALISATION — the whole question is driven into a single, well-defined step. We regard that as the paper's contribution: it converts 'can ropes do quantum mechanics?' into one precise, potentially decidable question.

2. The Local Baseline, Made Concrete

Before invoking non-locality we fix what a LOCAL rope model can do, so the gap non-locality must close is explicit. In a local model each detector outcome depends only on its own setting and a shared hidden orientation λ established at pair creation:

$$A(a,\lambda) = \text{sign}\cos(a-\lambda)\ ,\quad B(b,\lambda) = -\text{sign}\cos(b-\lambda) \quad (2.1)$$

Averaging over λ uniform on the circle gives, by direct computation, a TRIANGLE-WAVE correlation — linear in the angle — not the cosine, and a CHSH value that reaches but cannot exceed the classical bound:

Quantity	Local rope model	Quantum target
Correlation E(a,b)	triangle wave (linear in a−b)	−cos(a−b)
Max |E − (−cos)|	≈ 0.21 (at 45°)	0
CHSH S	2.00 (numerically 2.00)	2√2 ≈ 2.83

This is Bell's theorem made concrete in the rope picture: a local shared-orientation model is falsified by the same experiments that falsify all local hidden-variable theories. The task for non-locality is now sharp — turn the triangle wave into a cosine, and push CHSH from 2 toward 2√2, without letting the marginals carry a signal.

3. A Non-Local Update Model

The rope ontology of the foundations paper is specific: an entangled pair is a SINGLE connected topological configuration under a persistent global constraint, and measurement at one end UPDATES the joint state — constrains it, does not causally act on it. We build the minimal model with that structure. Measurement at A, with setting a, yields an outcome and updates the shared configuration; B's outcome is then drawn from the conditional distribution of the UPDATED joint state, which depends on both b and (through the update) a.

With the conditional probability that B is anti-correlated with A taken to be the squared spinor overlap of the two settings,

$$P(B = -A \mid a,\ b) = \cos^2((a-b)/2) \quad (3.1)$$

the model reproduces the quantum correlation and saturates the Tsirelson bound, verified by direct simulation:

Quantity	Non-local update model	Quantum target
Correlation E(a,b)	−cos(a−b) (max error 0.002)	−cos(a−b)
CHSH S	2.83 (numerically 2.826)	2√2 ≈ 2.828
Marginal ⟨A⟩, ⟨B⟩	0.00 (independent of the far setting)	0

The essential honesty flag, stated immediately. This model reaches the quantum result — but (3.1) was IMPOSED. Writing the conditional as the squared spinor overlap inserts the very quantum law we are trying to derive. Reproducing the cosine this way demonstrates CONSISTENCY (a non-local update CAN carry the quantum correlation) but is not, by itself, a DERIVATION from rope dynamics. Whether the rope ontology GENERATES (3.1) rather than merely permitting it is the question of §5 — and is where the honest wall stands.

4. No-Signalling Closes Cleanly

One part of the problem closes without qualification. In the update model the marginal outcome at each end is unaffected by the distant setting: ⟨A⟩ is independent of b and ⟨B⟩ is independent of a, both equal to zero within statistical error across all settings tested. The update changes only the JOINT distribution — the correlation — never the local marginals. There is therefore no channel by which choosing b could transmit information detectable at A: the non-locality is correlational, not causal, exactly as the foundations paper required.

What closes here

No-signalling holds identically for the non-local update rule, and does so independently of the unresolved angle-identification of §5. Whatever the resolution of that step, the rope update rule does not permit superluminal signalling. This is a genuine, unqualified success: the ontology's central consistency requirement — non-local correlation without signalling — is satisfied by an explicit mechanism.

5. Deriving the Conditional: The Attempt and the Wall

The real question is whether rope dynamics GENERATE the conditional (3.1). We attempt the derivation from the programme's own structural results and report precisely where it succeeds and where it stops.

5.1 *What the rope structure supplies*

Two ingredients come from the programme rather than from insertion. First, the two-strand state space is the Hopf bundle $S^3 \to S^2$, whose fibre structure is exactly the spin-½ spinor — a structural result of the programme's strand-linking analysis, not assumed here. Second, the measurement model is irreversible configuration-entropy increase ($S = k \ln|C|$), so an outcome's probability is set by the fraction of the shared configuration space compatible with it.

These two give more than one might expect. On a spinor, the squared overlap satisfies the exact half-angle identity

$$\cos^2(\theta/2) = (1 + \cos\theta)/2 \quad (5.1)$$

so the Born-rule form $\cos^2(\theta/2)$ is IDENTICAL to a probability that is LINEAR in the projection $\cos\theta$. In other words, once the state space is the spinor, one does not need to separately postulate 'square the amplitude': a configuration-fraction rule linear in the spin projection AUTOMATICALLY produces the Born $\cos^2$. The square is not an extra assumption on the spinor — it is linearity in the Bloch-sphere projection. This is a real, and slightly surprising, point in the programme's favour.

5.2 *The step that does not close*

The angle θ in (5.1) is the angle on the Hopf BASE sphere S^2 (the Bloch sphere), which is TWICE the angle in the spinor fibre. The detector settings a, b are physical laboratory angles. The quantum correlation $-\cos(a-b)$ requires that the laboratory angle difference map to the Bloch angle DIRECTLY, one-to-one ($\theta = a-b$), so that the projection $\cos\theta = \cos(a-b)$. The rope geometry, however, supplies BOTH angles — the base-sphere angle and the fibre angle, differing by a factor of two — and does not yet FORCE which one the measurement couples to.

This is not a vague gap; it is a single number. Parametrise the coupling by γ, with the detector seeing Bloch angle $\gamma(a-b)$. The CHSH value is then fixed ENTIRELY by γ:

Angle map γ	**Detector couples to**	**CHSH S**	**Verdict**
$\gamma = 1$	base-sphere (Bloch) angle	2.83	QUANTUM — exact Tsirelson
$\gamma = ½$	spinor fibre angle	2.39	sub-quantum
$\gamma = 2$	double angle	0.00	unphysical

So the entire viability of the rope model as a theory of quantum correlations reduces to one question: does the rope measurement dynamics couple the detector to the base-sphere angle, i.e. is $\gamma = 1$? At $\gamma = 1$ the model IS quantum mechanics — cosine, Tsirelson, no-signalling. Away from $\gamma = 1$ it is not. We could not derive $\gamma = 1$ from the rope dynamics; nor could we derive any other value. The dynamics as currently formulated leave γ undetermined.

6. Is the Update Model Secretly Local?

The foundations paper flagged a central trap: if a 'non-local' model's correlations are secretly reproducible by a pre-established common cause, it is really LOCAL and already falsified. We must check the update model against this.

It passes, because it exceeds the CHSH bound of 2 (reaching 2.83 at $\gamma = 1$). No local hidden-variable model can exceed 2; a model that reaches 2.83 is therefore genuinely non-local, not a disguised common cause. The non-locality enters specifically through the UPDATE: B's conditional distribution depends on a via the post-measurement joint state, which no pre-set λ alone can encode. So the model is honestly non-local where it needs to be — the gain over the local baseline of §2 is real, not bookkeeping.

A subtlety worth stating. That the model is genuinely non-local does not by itself make it CORRECT — it makes it the right KIND of theory. Correctness still hangs on $\gamma = 1$ (§5). The local-disguise check confirms we have not fooled ourselves into a local model wearing non-local clothing; it does not supply the missing angle identification.

7. What Would Close the Gap

Because the gap is now a single number, we can say concretely what would close it — which is more than an open problem usually offers:

1. A derivation, from the rope measurement dynamics, that the detector couples to the base-sphere (Bloch) angle — i.e. that the irreversible configuration-entropy measurement projects onto the S^2 base of the Hopf bundle, not its fibre. This would fix $\gamma = 1$ and complete the reproduction of quantum correlations.
2. Alternatively, a demonstration that the rope dynamics FORCE $\gamma \neq 1$ — which would be a genuine falsification, predicting a specific sub-quantum CHSH value (e.g. 2.39 at $\gamma = ½$) that experiment has already excluded. This outcome is equally decisive and equally valuable: it would rule the rope model out as a theory of quantum correlations.
3. Either way, the question is now EMPIRICALLY sharp: the rope model predicts a CHSH value that is a definite function of γ, and γ is fixed by the measurement dynamics. Pinning γ decides the matter.

This is the sense in which the partial result is progress. The foundations paper left 'reproduce the cosine' as an open programme; this paper reduces it to 'derive one angle identification,' with both possible answers ($\gamma = 1$ confirming, $\gamma \neq 1$ falsifying) sharply defined.

8. Honest Verdict

Where the rope model stands on quantum correlations

It is the right KIND of theory: non-local hidden-variable, genuinely exceeding the Bell bound, preserving no-signalling — all confirmed by explicit construction.

> It CAN carry the exact quantum correlation — cosine and Tsirelson — through a non-local joint update.
>
> It does NOT yet DERIVE that correlation: the update's conditional was imposed, and the one ingredient needed to generate it — coupling to the Bloch angle, $\gamma = 1$ — is not forced by the current dynamics.
>
> The viability question is therefore neither answered yes nor no; it is LOCALISED to a single angle-identification, with both resolutions empirically decisive.

We resist two temptations. We do not claim the reproduction of the cosine as a success of the programme — it used an imposed conditional. And we do not claim the failure to derive γ as a refutation — the model is the correct kind of theory and is one identification away from working. The accurate statement is the narrow one: the rope model is a consistent non-local ontology whose agreement with quantum mechanics hinges entirely on whether its measurement dynamics select the base-sphere angle, which remains to be shown.

9. Status of Each Result

Result	Status
Local rope model falsified (CHSH ≤ 2, triangle wave)	Established — direct computation
Non-local update reproduces $-\cos(a-b)$ and Tsirelson	Established — but with an IMPOSED conditional
No-signalling holds for the update rule	Established — marginals independent of far setting
Model is genuinely non-local (not disguised-local)	Established — exceeds CHSH = 2
Spinor state space from the Hopf structure	Inherited — structural result of the programme
Born $\cos^2$ = linearity in Bloch projection	Derived — half-angle identity (5.1)
Detector couples to Bloch angle ($\gamma = 1$)	OPEN — not forced by the dynamics; the whole viability turns on it
Rope dynamics uniquely generate the conditional	Open Problem — the central unmet step

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Addendum: The Measurement Problem Decomposed (QB-007, QB-008)
The single-detector side of the measurement problem — distinct from the two-party CHSH wall treated above, which is unchanged by everything below — has been decomposed into three labeled parts, two closed and one isolated with quantitative bounds. (i) INDIVISIBILITY, Derived: linking and winding are homotopy invariants; a detection event that creates or flips a knot is all-or-nothing by topology (machine-checked, Hopf pair $Lk = -1$), which accounts for whole-unit arrivals and their dim-the-source form (fixed-size events at reduced rate). (ii) THE SINGLE-SITE BORN RATE LAW, derived-in-structure in the weak-field limit: a nucleation site is a threshold (barrier-escape) system, and simulated Kramers dynamics under weak periodic driving gives a rate enhancement scaling as amplitude^1.7–2.1 against a pre-committed 2.0 ± 0.3 bar — threshold crossing responds to energy, so single-site click rates go as $|\psi|^2$, and twenty

thousand one-at-a-time draws rebuild the two-slit fringes at correlation 0.997. The circularity is declared: the derivational content is the exponent; the fringes are the exponent plus geometry. This parallels known semiclassical detection theory and is claimed as a registered decomposition within the rope framework, not as a novel result about detectors. Benchmark: benchmarks/bell/knot_nucleation_measurement.py.

(iii) THE ISOLATED CORE, registered as a negative with the discriminator quantified: anticorrelation. For any classical field driving independent threshold sites, Cauchy–Schwarz pins $g^2(0) \geq 1$ (coherent drive: exactly 1, machine-checked at 2×10^6 windows; fluctuating drive: bunched, ≈ 1.96), while the measured single-photon value is ≈ 0.18 (Grangier–Roger–Aspect 1986; modern values ≈ 0). Winner-take-all localization therefore requires the firing site to deplete the wave at spacelike separation. QB-008 corners that requirement: Bell-timing experiments (Salart et al. 2008; Yin et al. 2013) bound any preferred-frame influence at $v > 1.38\times10^4$ c for frames within $\sim10^{-3}$ c of Earth — a class containing the CMB rest frame, the mesh frame's natural identification — which via $v_L/c = \sqrt{K_L/K_T}$ demands a strand stretch-to-transverse stiffness ratio of at least 1.9×10^8; and the Bancal et al. (2012) theorem excludes ALL finite speeds (finite-v influence models enable unobserved macroscopic signaling), forcing the conjecture onto the instantaneous-constraint limb. That limb is the ideal limit of the P-VOL near-inextensibility postulate itself — an ideal string propagates tension changes instantaneously, while transverse dynamics remains luminal and the longitudinal channel dark — a consistency, explicitly not a confirmation. Status: the conjecture remains a Conjecture, now with one precisely specified limb, a quantified mechanical demand, and no middle ground; what would move it is a demonstration that instantaneous constraint propagation quantitatively produces $g^2 \approx 0$, which is nobody's result yet. Benchmark: benchmarks/bell/depletion_speed_bounds.py.

Appendix A. Dependency Graph

What closes (solid) and the single step that does not (dashed).

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  Foundations: rope entanglement => NON-LOCAL hidden variable  [established]
                                          |
                                          v
      local rope model  --->  CHSH <= 2, triangle wave  [FALSIFIED, solid]
                                          |
                           v  add non-local joint update
cosine + Tsirelson (2.83)      no-signalling holds  [solid, but conditional IMPOSED]
                  |                                     |
                  v                                     v
    genuinely non-local (>2)       marginals carry no signal  [both solid]
                                          |
                  v  derive the conditional from rope dynamics
     spinor state space (Hopf)  +  Born = linear-in-projection  [solid]
                                          |
                                          v
  - - - detector couples to Bloch angle, gamma = 1? - - -  [OPEN, dashed]
                                          |
                                          v
         gamma=1 => quantum   |   gamma!=1 => falsified  [decision point]
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Every solid link is established by direct computation; the single dashed link — the angle identification — carries the entire remaining question, with both outcomes empirically decisive.

Conclusion

We attempted the calculation the foundations paper deferred — reproducing the singlet correlation at the Tsirelson bound from rope dynamics while preserving no-signalling — and we report a partial result without dressing it as more. A local rope model is falsified, exactly as Bell requires. A non-local joint-update model, faithful to the rope ontology of a single connected configuration updated by measurement, reproduces the cosine, saturates Tsirelson, preserves no-signalling, and is genuinely non-local rather than a disguised common cause. But it reaches the quantum value through a conditional that we imposed rather than derived, and the one ingredient that would generate that conditional — the measurement coupling to the Hopf base-sphere angle, $\gamma = 1$ — is not forced by the current dynamics. The rope model is thus shown to be the correct KIND of theory for quantum correlations, and one precise identification away from reproducing them. That identification, and whether the rope dynamics supply it, is the sharp and decidable question this paper leaves open. Reporting it as open, rather than claiming a derivation we did not achieve, is the point.

Programme credit: the core Rope Hypothesis is due to Bill Gaede. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.

Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

• **QB-001 [Derived]:** Local rope model gives triangle wave, CHSH ≤ 2 [benchmark: benchmarks/bell/nonlocal_correlation.py]